SCOTT LUCCHINI

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I study gas dynamics and its role in galaxy formation and evolution. Using cosmological and idealized simulations I have studied the Magellanic Stream and the broader circumgalactic medium with a focus on making mock observations to compare with data.

EDUCATION

2023 **PhD in Physics** University of Wisconsin–Madison
Thesis: The Magellanic Corona and its Role in the Evolution of the Magellanic Stream
Advisor: Elena D'Onghia, Professor of Astronomy

2017 **M.Sc. with Distinction in Theoretical Physics** University of Edinburgh
Thesis: The Quantum Nature of Self-Dual Yang-Mills Theory
Advisor: Donal O'Connell, Professor of Physics and Astronomy

2014 **B.S. in Physics and Astronomy** University of Rochester

RESEARCH AND PROFESSIONAL EXPERIENCE

2023– **ITC Postdoctoral Fellow** Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA
Advisors: Lars Hernquist, Professor of Astronomy, Charlie Conroy, Professor of Astronomy

2014–16 **Software Developer** Heretto, Rochester, NY

REFEREED PUBLICATIONS – ADS

24 total, 12 first author, 2 second author, 10 third+ author with significant contributions *mentored students are underlined

Selected Publications

14. ENhanced Galactic Atmospheres With Arepo: Resolving the CGM at 200 pc with the ENGAWA Simulations
Lucchini, S., Abramson, C., Hummels, C., Conroy, C., Hernquist, L., & Smith, A. *ApJ*, accepted. (2026) arXiv:2603.05584
13. Invisible Accretion: Ionized Envelopes of TNG50 HVCs can Sustain Star Formation
Lucchini, S., Han, J., Hernquist, L., Conroy, C., & Fox, A. J. *ApJ*, 990, 118. (2025) 10.3847/1538-4357/adf3b3
12. The Magellanic Corona as the key to the formation of the Magellanic Stream
Lucchini, S., D'Onghia, E., Fox, A. J., Bustard, C., Bland-Hawthorn, J., & Zweibel, E. *Nature*, 585, 203. (2020) 10.1038/s41586-020-2663-4 arXiv:2009.04368

Additional First and Second Author Publications

11. A New Model for the Milky Way Halo Dispersion Measure with the ENGAWA Simulations: Low DMs and Large Anisotropy
Lucchini, S., Connor, L., McCarty, S., Konietzka, R. M. *ApJ*, in review. (2026) arXiv:2607.20601
10. The LMC Corona Favors a First Passage
Lucchini, S., Han, J., Mishra, S., Fox, A. J. *ApJ*, 1002, 14. (2026) 10.3847/1538-4357/ae5bc3
9. Threading the Magellanic Needle: Hypervelocity Stars Trace the Past Location of the LMC
Lucchini, S. & Han, J. *ApJL*, 993, L10. (2025) 10.3847/2041-8213/ae109d
8. On the Origin of High-velocity Clouds in the Galaxy
Lucchini, S., Han, J., Hernquist, L., & Conroy, C. *ApJ*, 974, 105. (2024) 10.3847/1538-4357/ad6dde
7. Following the tidal trail: a history of modeling the Magellanic Stream
Lucchini, S. *Astrophys Space Sci*, 369, 114. (2024) 10.1007/s10509-024-04377-5 PDF
6. Properties of the Magellanic Corona
Lucchini, S., D'Onghia, E., & Fox, A. J. *ApJ*, 967, 16. (2024) 10.3847/1538-4357/ad3c3b

5. The Milky Way Bar Pattern Speed using Hercules and Gaia DR3
Lucchini, S., D'Onghia, E., & Aguerri, J. A. L. *MNRAS*, 531, L14. (2024) 10.1093/mnrasl/slae024
4. Moving Groups Across Galactocentric Radius with Gaia DR3
Lucchini, S., Pellett, E., D'Onghia, E., & Aguerri, J. *MNRAS*, 519, 432. (2023) 10.1093/mnras/stac3519
3. First evidence of a stripped star cluster from the Small Magellanic Cloud
Piatti, A. E. & **Lucchini, S.** *MNRAS*, 515, 4005. (2022) 10.1093/mnras/stac1980
2. The Magellanic Stream at 20 kpc: A New Orbital History for the Magellanic Clouds
Lucchini, S., D'Onghia, E., & Fox, A. J. *ApJL*, 921, L36. (2021) 10.3847/2041-8213/ac3338
1. Using kinematic properties of pre-planetary nebulae to constrain engine paradigms
Blackman, E. G. & **Lucchini, S.** *MNRAS*, 440, L16. (2014) 10.1093/mnrasl/slu001

Third+ Author Publications

10. Tracing Winds through the Fog: A Comprehensive Survey of the LMC's Galactic Outflows with ULLYSES
Poudel, S., Horton, A., et al., incl **Lucchini, S.** *ApJ*, accepted. (2026) arXiv:2609.32137
Contribution: Member of the ULLYSES proposal team
9. The Legacy of N11: Probing Galactic Winds from an Evolved Star Formation Region Using ULLYSES
Horton, A., Poudel, S., et al., incl **Lucchini, S.** *ApJ*, 1006, 198. (2026) 10.3847/1538-4357/ae8157
Contribution: Member of the ULLYSES proposal team
8. Secular Attrition of Classical Bulges by Stellar Bars
McClure, R. L., Géron, T., et al., incl **Lucchini, S.** *ApJ*, in review. (2025) arXiv:2506.09150
Contribution: Assisted with Bayesian KDE analysis and provided general feedback
7. New Interpretation for the Orientation of the LMC's Gaseous Arms B and E using ULLYSES
Horton, A., Poudel, S., et al., incl **Lucchini, S.** *ApJ*, 993, 195. (2025) 10.3847/1538-4357/ae00be
Contribution: Mock observations of LMC and provided Figures 10, 11
6. A Preinfall Magellanic Analog–Corona and Stream Formation in the HESTIA Cosmological Simulations
Chisholm, R., D'Onghia, E., et al., incl **Lucchini, S.** *ApJ*, 993, 67. (2025) 10.3847/1538-4357/ae0ccc
Contribution: Comparisons with isolated LMC models and analysis techniques
5. The Distance to the Magellanic Stream: Constraints from Optical Absorption along Stellar Sightlines
Mishra, S., Fox, A. J., et al., incl **Lucchini, S.** *ApJ*, 984, 104. (2025) 10.3847/1538-4357/adc68a
Contribution: Determined selection of stars and contributed models
4. The Gaseous Blowout of the 30 Doradus Starburst Region in the LMC
Poudel, S., Horton, A., et al., incl **Lucchini, S.** *ApJ*, 984, 161. (2025) 10.3847/1538-4357/adc099
Contribution: Comparison with simulations and provided Figures 18, 19
3. Hypervelocity Stars Trace a Supermassive Black Hole in the Large Magellanic Cloud
Han, J., El-Badry, K., **Lucchini, S.**, et al. *ApJ*, 982, 188. (2025) 10.3847/1538-4357/adb967
Contribution: Assisted with orbital histories for LMC/SMC/MW system
2. The Truncated Circumgalactic Medium of the Large Magellanic Cloud
Mishra, S., Fox, A. J., et al., incl **Lucchini, S.** *ApJL*, 976, L28. (2024) 10.3847/2041-8213/ad8b9d
Contribution: Comparison with simulations and ram pressure calculations
1. Observations of a Magellanic Corona
Krishnarao, D., Fox, A., et al., incl **Lucchini S.** *Nature*, 609, 915. (2022) 10.1038/s41586-022-05090-5
Contribution: Provided properties of predicted Magellanic Corona from simulations

AWARDS & GRANTS

Research Awards

Aug. 2024	International Astronomical Union PhD Prize	
Jun. 2022	NASA Wisconsin Space Grant Consortium Research Fellowship	$5k
Apr. 2022	AAS Division on Dynamical Astronomy 2022 Duncombe Prize	$600
Jul. 2021	Karl Guthe Jansky and Alice Knapp Jansky Scholarship, UW–Madison Physics Dept.	$3k
Apr. 2021	NASA Wisconsin Space Grant Consortium Research Fellowship	$5k
Dec. 2020	Stebbins Award, UW–Madison Astronomy Dept.	$2k
May 2020	Karl Guthe Jansky and Alice Knapp Jansky Scholarship, UW–Madison Physics Dept.	$3k

Teaching Awards

Apr. 2022	Joseph R. Dillinger Award for Teaching Excellence	UW–Madison Physics Dept.
May 2020	Best TA Award (General Physics – PHY 104, F19)	UW–Madison Physics Dept.
May 2018	Best TA Award (General Physics – PHY 207, F17)	UW–Madison Physics Dept.
May 2013	Undergraduate Teaching Award	University of Rochester Physics Dept.

Travel Awards

Apr. 2022	Division on Dynamical Astronomy Travel Award	$175
2020, 2022	UW Student Research Grants	$1200

Awarded Observational Proposals

Aug. 2025	**Magellan/LLAMAS Fall 2025** Co-I: Introducing $(MC)^2$: The Magellanic Clouds Magellan Campaign	4 nights
Jul. 2025	**Cycle 33 Hubble Space Telescope Archival Research Proposal** Co-I: "Ionized Gas around the Small Magellanic Cloud"	
Jul. 2025	**Cycle 33 Hubble Space Telescope GO Research Proposal** Co-I: "Where is the central black hole in the LMC?"	7 orbits
Jul. 2025	**Cycle 33 Hubble Space Telescope GO Research Proposal** Co-I: "Unveiling the Circumgalactic Medium of M33"	30 orbits
Jul. 2025	**Cycle 33 Hubble Space Telescope GO Research Proposal** Co-I: "The Fate of the Leading Arm of the Magellanic Stream"	22 orbits
Jul. 2024	**Cycle 32 Hubble Space Telescope GO Research Proposal** Co-I: "Probing the front-side of the Circumgalactic Medium of the Large Magellanic Cloud"	29 orbits
Feb. 2023	**Period 111 Very Large Telescope Proposal** Co-I: "The Distance to the Magellanic Stream"	26h
Jun. 2022	**Cycle 30 Hubble Space Telescope Archival Research Proposal** Co-I: "The Cool CGM of the Large Magellanic Cloud"	
Jun. 2021	**Cycle 29 Hubble Space Telescope Archival Research Proposal** Co-I: "The LMC's Galactic Wind through the Eyes of ULLYSES"	$110k
May 2020	**Cycle 28 Hubble Space Telescope Archival Research Proposal** Co-I: "Searching for the LMC Corona: The missing element for the formation of the Magellanic Stream"	$276k

PRESENTATIONS

Selected Invited Talks & Colloquia

Jan. 2026	Confronting Simulations with the Real Universe Conference	Napa, CA
Nov. 2025	Weber State Physics & Astronomy Colloquium	Ogden, UT
Oct. 2024	Decade of Discovery: Celebrating 10 Years of the Illustris Project	Gargonza, Italy
Aug. 2024	IAU GA – Division J PhD Prize Seminar [Recording Link]	Cape Town, South Africa
Apr. 2024	Via Science Meeting	Cambridge, MA
Nov. 2023	Univ. Vienna Institute for Astrophysics Colloquium	Virtual
Jun. 2022	GBO HVCs Workshop Invited Talk	Green Bank, WV

Selected Seminars

Mar. 2026	UC Santa Barbara Astro Lunch Talk	Santa Barbara, CA
Mar. 2026	Caltech Tea Talk	Pasadena, CA
Nov. 2025	MIT Monday Afternoon Talks	Cambridge, MA
Oct. 2025	Yale Cosmology Seminar	New Haven, CT
Sept. 2025	Stanford KIPAC Seminar	Stanford, CA
Sept. 2025	UC Santa Cruz Friday Lunch Astrophysics Seminar Hour	Santa Cruz, CA
Sept. 2025	UC Berkeley Astrophysics Lunch Seminar	Berkeley, CA
Feb. 2025	CfA ITC Luncheon Seminar [Recording Link]	Cambridge, MA
Sept. 2023	CfA ITC Luncheon Seminar [Recording Link]	Cambridge, MA
Apr. 2023	CU Boulder Astrophysical & Planetary Sciences Friday Lunch Seminar	Boulder, CO
Mar. 2023	Univ. Maryland Center for Theory and Computation Seminar	College Park, MD
Oct. 2022	CCA Galaxy Evolution Group Meeting Seminar	New York, NY
Oct. 2022	Columbia University Seminar	New York, NY

Oct. 2022	Space Telescope Science Institute Low Density Universe Seminar	Baltimore, MD
Jul. 2022	Madison Astronomical Society Public Talk [Recording Link]	Madison, WI
Mar. 2022	Space Telescope/Johns Hopkins Galaxies and AGN Seminar	Virtual

Selected Contributed Talks

Jul. 2025	Cosmic Ecosystems [Recording Link]	Waterloo, ON
May 2025	XMC II: Clouds Over Yellowstone [Recording Link]	Bozeman, MT
Jan. 2025	245th American Astronomical Society Meeting	National Harbor, MD
Aug. 2024	Multiphase Madness CGM Conference [Recording Link]	Cambridge, MA
Jun. 2024	244th American Astronomical Society Meeting	Madison, WI
Feb. 2024	XMC Workshop: Milky Clouds Over Manhattan	New York, NY
Apr. 2022	53rd Division on Dynamical Astronomy of the AAS Annual Meeting	New York, NY
Sept. 2019	A synoptic view of the Magellanic Clouds: VMC, Gaia and beyond	Garching, Germany

TEACHING & ADVISING EXPERIENCE

Co-Instructor – UW–Madison

Taught five lectures over the eight-week session including active learning components (clicker questions) and live demonstrations. Rewrote two lectures including pre-lecture and bridge material and wrote new problems for exams and discussion worksheets.

General Physics – algebra-based mechanics (PHY 103), Su22 — 50 students

Teaching Assistant – UW–Madison

Created worksheets and quizzes each week. Led discussion sections and labs where group work and interactive learning was encouraged. Graded problem sets and exams.

General Physics – algebra-based mechanics (PHY 103), S20	75 students (~400 total)
General Physics – algebra-based E&M (PHY 104), F19	75 students (~400 total)
General Physics – calculus-based E&M (PHY 202), Su18, F18, S19, Su19	50 students (~500 total)
General Physics – calculus-based mechanics (PHY 207), F17, S18	50 students (~300 total)
Electromagnetic Fields (PHY 323), S21	20 students
Theoretical Physics – Dynamics (PHY 711), F21	30 graduate students

Guest Lecturer – UW–Madison

Our Exploration of the Solar System (AST 104), S22
General Physics – algebra-based mechanics (PHY 103), review lecture S20

Teaching Intern – University of Rochester

Led labs and workshop sessions and graded coursework and exams for a variety of courses.

Advising Experience

Two UW–Madison undergraduate students from 2020–2023 developing and using the MGWave open-source python wavelet transformation code. Resulted in publication of *Moving Groups Across Galactocentric Radius with Gaia DR3*.

Two Caltech undergraduate students during Summer 2024 and Summer 2025 SURF programs. Population analysis of cold gas in the circumgalactic medium and study of impact of numerical resolution.

SERVICE AND ORGANIZATIONS

Chair of organizing committee for Multiphase Madness CGM conference	Aug. 2024
ITC Luncheon talk series coordinator	since 2023
UW Physics Outreach, Museum, Web Content & Events committee member	2022–2023

Member – American Astronomical Society, GASKAP
Reviewer – AAS Journals, MNRAS, Nature Astronomy

REFERENCES

Elena D'Onghia (PhD advisor)	University of Wisconsin–Madison	edonghia@astro.wisc.edu
Lars Hernquist	Harvard	lhernquist@cfa.harvard.edu
Charlie Conroy	Harvard	cconroy@cfa.harvard.edu
Alyssa Goodman	Harvard	agoodman@cfa.harvard.edu
Andy Fox	Space Telescope Science Institute	afox@stsci.edu
Bob Benjamin	University of Wisconsin–Whitewater	benjamir@uww.edu

MEDIA AND PRESS

The LMC Corona Favors a First Passage
Lucchini, S., et al. *ApJ*, 1002, 14. (2026)
- Phys.org: "Is the Large Magellanic Cloud a first-time visitor?"

Hypervelocity Stars Trace a Supermassive Black Hole in the Large Magellanic Cloud
Han, J., El-Badry, K., Lucchini, S., et al. *ApJ*, 982, 188. (2025)
- CfA: "Runaway Stars Reveal Hidden Black Hole In Milky Way's Nearest Neighbor"
- NPR: "That galaxy next door? It's home to a monster black hole"
- Forbes: "Supermassive Black Hole Lurks In Nearby Large Magellanic Cloud"
- Smithsonian: "Astronomers Discover Evidence of a Stealthy Supermassive Black Hole…"
- Reuters: "Fast-moving stars reveal supermassive black hole inside nearby galaxy"
- Phys.org, New Scientist, Discover Magazine, Space.com, Sci News

The Truncated Circumgalactic Medium of the Large Magellanic Cloud
Mishra, S., et al., incl. Lucchini, S. *ApJL*, 976, L28. (2024)
- NASA: "NASA's Hubble Sees Aftermath of Galaxy's Scrape with Milky Way"
- Universe Today, SciTechDaily

Observations of a Magellanic Corona
Krishnarao, D., et al., incl. Lucchini, S. *Nature*, 609, 915. (2022)
- NASA: "Using Quasars to Map the Magellanic Corona"
- UW–Madison: "Decades of work at UW–Madison underpin discovery of corona…"
- Hubblesite: "Hubble Detects Protective Shield Defending a Pair of Dwarf Galaxies"
- Space.com, Sci News, CNET, Universe Magazine

The Magellanic Stream at 20 kpc: A New Orbital History for the Magellanic Clouds
Lucchini, S., et al. *ApJL*, 921, L36. (2021)
- UW–Madison: "Magellanic Stream arcing over Milky Way may be five times closer than previously thought"
- Phys.org, Science Alert, Live Science

The Magellanic Corona as the key to the formation of the Magellanic Stream
Lucchini, S., et al. *Nature*, 585, 203. (2020)
- UW–Madison: "Huge Halo of Warm Gas around Magellanic Clouds is Key to Formation of Magellanic Stream"
- Univ. Sydney: "How the Milky Way stole an enormous gas halo from our dwarf neighbours"
- Nature: "Galactic coronae"
- Phys.org, Universe Today, CNET, VICE, Cosmos Magazine, Sci News, Science Daily, Science Alert